Exp: 7.

Date: 24-9-26. Sliding window.

AIM:

Write a program to implement flow control at datalink layer using sliding window protocol simulate flow of frames. from one node to another.

Sender.py

```
import time.
import os
Sender_file = Sender_Buffer.txt
Receiver_file = Receiver_Buffer.txt.

def create_sender_buffer(frames):
"write the current window frames to sender Buffer."""
with open(Sender_file, "w") as file:
for frame_no, data in frames:
file.write(f"Frame No: {frame_no}, Data {data}\n")

def read_acks():
"read Acks/NACks from Receiver_Buff."""
if not os.path.exsits(Receiver_File):
return None.
with open(Receiver_file, "r") as file:
response = file().read().strip()
if response.startswith("ACK:"):
return "ACK", int(response.split(":")[1].strip())
if response.startswith("Nack:"):
return "nack", int(response.split(":")[1].strip())
```

return none..

Ex.No: 7

Date: 24-9-26

Sliding window.

AIM:

Write a program to implement flow control at datalink layer using sliding window protocal. simulate flow of frams, from one node to another.

Sender.py:

```
import time.
window = int (input ("Enter window size:"))
message = input ("Enter message: ")
frames = [(i+1, ch) for i, ch in enumerate(message)]
start=0
while start < len (frams):
end = min (start + window, len (frames))
current = frams [ start : end ]
with open ("sender_Buffer.txt", "w") as f:
    for no, data in current:
        print (f"Sender : Frame {no} → {data}")
        f. write (f" {no}, {data} \n")
time . sleep (3)
with open ("Receiver Buffer. txt, "r") as f:
    response = f. read (). strip()
```

```
if response.startswith("ack"):
int( response.startswith("Awk"):
int ( response.split(":")[1])
    if ack == current[-1][0]:
    print(f"sender: ack {ack} received")
    start = end
else:
    print("sender: wron ack")
elif
    response.starts with("nack"):
    nack = int(response.split(":")[1])
print(f"sender: nack {nack} received")
    start = nack-1;
```

receiver.py

```
    expected = 1
while True:
    input("press Enter to read sender-Buff...")
with open["senderBuffer.txt","r") as f:
    frames = f.readlines()
```

```
    for line in frames:
        no, data = line.strip().split(",")
        no = int(no)
    print(f"Receiver: Wrong frame! Expected {Expected}")
    open("Receiver_Buffer.txt", "w") as f:
        f.write(f"Nack: {expected}")
        break

else:
    with open("Receiver_Buffer.txt", "w") as f:
    f.write(f"Nack: {expected}")
    break;

else:
with open(Receiver_Buffer.txt", "w") as f:

    f.write(f"Ack: {ack}")
    print(f"Receiver: ACK {ACK} sent")
```

output:

```
Enter window size : 3
Enter message : Hellow.

Sender: Frame 1 -> H
Sender: Frame 2 -> E
Sender: Frame 3 -> L
Press enter to read Sender_Buffer....
Receiver: frame 1 -> H
Receiver: frame 2 -> E
receiver: frame 3 -> L
Receiver: Ack 3 sent.
```

Sender : Ack 3 received.
Sender : Frame 4 → L
Sender : Frame 5 → 0.

Receiver
press enter to read Sender-Buffer....
Receiver : frame 4 → L
Receiver : frame 5 → 0
Receiver : Ack 5 sent.

Sender : Ack 5 received.

Result:
thus the sliding window protocal was successfully implemented using frames, Ack/NACK and retransmission for flow control and Error handling.